

Section A: Pure Mathematics [40 marks]

1 On a single Argand diagram, sketch the following loci.

(i) $|z - 2 - 2i| = 1,$

(ii) $\left| z - \frac{5}{2} - 2i \right| = \frac{1}{2},$

(iii) $\arg\left(z - \frac{5}{2} - 2i\right) = \pi - \tan^{-1} 2.$ [4]

(a) The complex number w is represented by the point of intersection of the loci in parts (i) and (iii). Find w , in the form $x + iy$, where x and y are real. [2]

(b) Given that $|z - 2 - 2i| \leq 1$ and $\left| z - \frac{5}{2} - 2i \right| \geq \frac{1}{2}$, shade the region satisfying $\arg\left(z - \frac{5}{2} - 2i\right) \leq \pi - \tan^{-1} 2.$ [2]

2 The function f is defined by

$$f: x \mapsto \sqrt{x+2}, \quad x \geq -2.$$

(i) Explain why the function f^2 may be defined, where f^2 refers to the composite function ff . Define, in a similar form, the composite function f^2 and state its range exactly. [4]

(ii) Sketch, on a single diagram, the graphs of $y = f(x)$, $y = f^2(x)$ and $y = x$. [2]

The sequence of numbers $\{v_n\}$ is defined by

$$v_{n+1} = f(v_n), \quad n \geq 1.$$

(iii) Given that the sequence converges to a number v , find the value of v . [2]

(iv) Given that the composite function f^n may be defined for all positive integers n , state the range of f^n as $n \rightarrow \infty$. [1]

- 3** A company manufactures closed hollow cylindrical cans of cross-sectional radius r cm and height h cm.
- (i) A can is made of two different materials. Its top and base cost 0.09 cents per cm^2 and its curved surface costs 0.06 cents per cm^2 to manufacture. Show that the radius of the cheapest can of volume 300 cm^3 is $\sqrt[3]{\frac{a}{\pi}}$, where a is a constant to be determined. [5]
- (ii) At the beginning of an experiment, a can is filled with $500\pi \text{ cm}^3$ of liquid. Due to a leakage, the height of the liquid is decreasing at a constant rate of 0.02 cm s^{-1} . Given $r = 2h$, find the rate at which the volume of the liquid is decreasing 2 minutes after the start of the experiment. [6]
- (Area of curved surface of cylinder = $2\pi rh$)

4 In this question, give each of your answers in exact form.

The plane Π have equation $\mathbf{r} = (2 + \lambda - 3\mu)\mathbf{i} + (3 - 2\lambda)\mathbf{j} + (\mu - 2)\mathbf{k}$ where $\lambda, \mu \in \mathbb{R}$ and the point A has position vector $2\mathbf{j}$.

- (i) Express the equation of Π in the form $\mathbf{r} \cdot \mathbf{n} = p$ where $\mathbf{n}$ is a vector perpendicular to Π . [3]
- (ii) Find the position vector of the foot of perpendicular, B , from A to Π .
Deduce the perpendicular distance from A to Π . [5]
- (iii) By using a vector product, find the length of projection of $\vec{OA}$ on Π . [2]
- (iv) By using (ii) and (iii), find the area of triangle OAB . [2]

Section B: Statistics [60 marks]

- 5 Four boys and four girls are to be seated in a round table with eight seats.
Find the number of possible seating arrangements in which
- (i) no two boys are seated next to each other, [2]
 - (ii) all the seats are numbered and a boy A and a girl B are to be seated next to each other. [3]
- 6 (a) Two events A and B are such that $P(A) = 0.2$ and $P(A \cup B) = 0.6$.
Find $P(B)$ if
- (i) A and B are mutually exclusive, [1]
 - (ii) A and B are independent. [2]
- (b) In 2008, an insurance company classifies 10% of their car policy holders as “good risks”, 60% as “average risks” and 30% as “bad risks”. Their statistical database in 2008 has shown that of those classified as “good risks”, only 1% were involved in at least one accident whereas of those classified as “bad risks”, 25% were involved in at least one accident. As for those classified as “average risks”, 15% were involved in at least one accident.
Find the probability that
- (i) a randomly chosen car policy holder was involved in at least one accident in 2008. [2]
 - (ii) a randomly chosen car policy holder was classified as “good risks” if the car policy holder was not involved in any accidents in 2008. [3]
- 7 (a) The random variable X is such that $P(X \geq 1) = 0.96$.
88 independent observations of X are taken.
By using a suitable approximation, find the probability that at most 82 out of the 88 observations of X are greater or equal to 1. [4]
- (b) A scientist examined some pond water under a microscope and found that, on the average, there are 4 larvae in every 10ml of the pond water.
- (i) Find the probability that a random sample of 5ml of the pond water contains less than 3 larvae. [2]
- 6 test-tubes are each filled with 5 ml of the pond water.
- (ii) By using a suitable approximation, find the probability that the 6 test-tubes will contain more than 13 larvae altogether. [4]

- 8** The random variable X is normally distributed with mean 20 and variance σ^2 .
- (i) Given that $P(12 < X < k) = P(X > k) = 0.42065$, show that $\sigma = 8$, correct to the nearest whole number, and find the value of the constant k . **[4]**

X is related to a normal random variable Y by the formula $X = Y_1 + Y_2 - 4$, where Y_1 and Y_2 are two independent observations of Y . Another random variable W , where W and Y are independent, is normally distributed with mean 25 and variance 16.

- (ii) Using $\sigma = 8$, find $E(Y)$ and $\text{Var}(Y)$, and hence find $P(W > 2Y)$. **[5]**
- 9** In a certain institution, each Co-Curricular Activity (CCA) has exactly one student leader. The following table shows all the 150 student leaders from the institution for each of the four CCA types.

Sports	Clubs	Performing Arts	Uniform Groups
48	60	33	9

From these 150 student leaders, a random sample of 50 student leaders is to be chosen to take part in a survey to gather feedback on their CCA experiences.

- (i) Describe briefly how a stratified random sample of 50 student leaders may be chosen. **[2]**
- (ii) Give one reason why a stratified random sample might be preferable than a simple random sample in this context. **[1]**

From the above survey, the number of hours each student leader spent on CCA per week, x , was obtained. The summarised data are as follows:

$$\sum(x-5) = 32 \quad \text{and} \quad \sum(x-5)^2 = 400.$$

- (iii) Find the unbiased estimates of the population mean and population variance of the number of hours each student leader spent on CCA per week. **[2]**
- (iv) Based on the above sample, estimate the probability that the average number of hours each student leader spent on CCA per week exceeds 6 hours. **[3]**

- 10** The manufacturers of an electric kettle claim that their kettle will heat 2 litres of water from 25°C to 100°C in, on average, less than 440 seconds. In order to test the claim, 12 randomly chosen kettles are bought and the time, in seconds, to heat 2 litres of water from 25°C to 100°C are measured. The results are as follows:

419 447 371 430 436 442 433 421 445 422 438 443

- (i) Assuming that the time taken to heat 2 litres of water from 25°C to 100°C is normally distributed, test the validity of the manufacturers' claim at the 5% level of significance. [4]

It was later found that the population standard deviation of the time taken to heat 2 litres of water from 25°C to 100°C is known and the same data is used to test whether the manufacturers' claim is valid at the 5% level of significance.

- (ii) Under the same assumption made in (i), determine whether a z-test or a t-test is the correct test to be used, justifying your answer. [2]
- (iii) Suppose the correct test in (ii) does not reject the null hypothesis, determine whether it is possible for the incorrect test in (ii) to reject the null hypothesis at the 5% level of significance, explaining your answer clearly. [2]
- 11** 10 randomly selected students were asked to indicate the number of hours they spent studying Mathematics, x , during the March Holiday. Their corresponding Term 2 Mathematics Common Test score, out of 100 marks, y , were recorded as follows:

Student	1	2	3	4	5	6	7	8	9	10
x	3.8	2.4	3.0	7.2	9.4	3.6	8.2	2.6	2.0	4.8
y	47	19	30	78	84	38	76	23	10	64

- (i) Find the equations of the least squares regression line of y on x and that of x on y . Find also the product moment correlation coefficient for the data. [3]
- (ii) Draw a scatter plot of y against x , together with the two regression lines found in (i), showing clearly the coordinates of the point of intersection of the two regression lines on your scatter plot. [4]
- (iii) Use the appropriate least squares regression line found in (i) to estimate the Term 2 Mathematics Common Test score of a student who spent 12 hours studying Mathematics during the March Holiday. Comment on the reliability of this estimate. [2]
- (iv) Find the equation of the least squares regression line of y on $\frac{1}{x}$ and use it to estimate the Term 2 Mathematics Common Test score of the student in (iii). Give a reason why this is a better estimate than that found in (iii). [3]

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